HAADS • SIH 26050

HIGH-ALTITUDE RUGGEDIZED ANTI-DRONE DETECTION AND PRECISION TRACKING SYSTEM

with AI-Based Environmental Compensation

Concept reference image retained from the source project document.

PROJECT IDENTITY

Problem Statement: High-Altitude Performance Optimization and Robust Design of Anti-Drone System • Problem Statement ID: 26050 • Revision 1.0

Prepared: September 2026

Document Control & Engineering Status

Field	Value
Project	HAADS — High-Altitude Anti-Drone System
SIH Problem Statement	High-Altitude Performance Optimization and Robust Design of Anti-Drone System
PS ID	26050
Primary innovation	Environmental-aware adaptive sensing and precision tracking
Current software prototype	Python + Streamlit + YOLO26n general object detection + environmental simulation + compensation + health monitoring
Hardware simulation	Wokwi ESP32 simulation with BME280, MPU6050, pan/tilt servos and environmental input controls
Physical hardware status	Not yet treated as fully deployed hardware; future prototype architecture remains proposed
RF scope	Receive-only monitoring concept; no RF jamming, disruption or interception
Response scope	Alert, operator notification, tracking handoff and simulated authorized workflow only

IMPORTANT CLAIMS POLICY

The source document contains illustrative values such as detection confidence, tracking error, system health and mission-readiness percentages. These are not presented here as measured field results unless independently validated. The current vision prototype is described as general object detection; drone-specific detection requires a suitable drone-trained/fine-tuned model and dataset.

Contents

The Word table of contents field below is intended to be updated in Microsoft Word using References → Update Table. The numbered sections are kept stable so the document can also be navigated manually.

1. Executive Summary

HAADS is an environmental-aware sensing, tracking and system-health architecture intended to maintain dependable target detection and precision pan-tilt tracking when high-altitude operating conditions degrade conventional electronic, mechanical and power-system performance.

The central engineering idea is to measure environmental state — temperature, pressure, humidity, wind and vibration — and feed that state into compensation and health-monitoring logic. Rather than treating environmental effects as external disturbances, the system uses them as control inputs for adaptive tracking, sensor correction, thermal management and performance assessment.

The proposed system combines an EO camera and edge AI pipeline with environmental sensing, a real-time control layer, pan-tilt actuation, thermal control, battery monitoring and a ground-control dashboard. The academic prototype is intentionally simulation-first: software behavior and embedded control concepts can be demonstrated before physical hardware is manufactured.

CORE VALUE PROPOSITION

Environment → estimate disturbance/degradation → compensate → stabilize → monitor health → inform operator.

2. Problem Statement & Context

2.1 Operational problem

An anti-drone system is commonly framed around four functions: Detect → Identify → Track → Respond. At high altitude, environmental conditions can reduce the reliability of sensing, actuation, power delivery and electronics.

- Extreme cold
- Low atmospheric pressure
- Strong winds
- Dust and snow
- Reduced battery performance
- RF performance variation
- Mechanical stiffness
- Sensor drift

The source document defines the main objective as monitoring environmental conditions and automatically compensating for their effects on detection, tracking, stabilization and system health. ṆfileciteÖturn5file0ÖL10-L25Ộ

2.2 Engineering interpretation

Condition	Potential system effect	HAADS response
Low temperature	Cable stiffness, sensor bias, battery degradation, electronics outside preferred operating range	Temperature-aware compensation, calibration, thermal management, health warnings
Strong wind	Pan-tilt disturbance and pointing error	IMU + wind estimate + encoder feedback + adaptive control
Low pressure / low air density	Changes in environmental state and cooling effectiveness	Pressure monitoring and environmental state model; thermal design accounts for reduced convective cooling
Dust / snow / moisture	Ingress and contamination risk	Rugged enclosure, sealed connectors, protective coating and maintenance strategy
Vibration	Tracking noise, sensor disturbance, mechanical stress	IMU monitoring, vibration isolation and health monitoring

3. Objectives, Requirements & Success Criteria

3.1 Primary objectives

1. Continuously monitor environmental and system-health parameters.
2. Detect and track visible targets using an edge-AI vision pipeline.
3. Estimate environmental effects on tracking and system health.
4. Apply temperature-, wind- and sensor-aware compensation.
5. Maintain stable pan-tilt behavior using feedback.
6. Expose system state through a live operator dashboard.
7. Provide a simulation-first path to hardware validation.

3.2 Functional requirements

ID	Requirement	Verification approach
FR-01	Camera/video input available to AI pipeline	Live stream / test video
FR-02	Object detections include location and confidence	Bounding boxes and inference output

ID	Requirement	Verification approach
FR-03	Tracking maintains target state across frames	Tracker continuity test
FR-04	Environmental variables update continuously	Simulator/Wokwi telemetry
FR-05	Compensation state changes with environmental conditions	Scenario tests
FR-06	Health monitor reports subsystem state	Dashboard fault/status test
FR-07	Pan/tilt commands respond to target error	Simulation or servo test
FR-08	System distinguishes prototype facts from future capabilities	Documentation review

3.3 Engineering success criteria

For the student prototype, success should be demonstrated by repeatable behavior rather than invented absolute percentages. A strong demonstration is a before/after comparison under controlled simulated environmental conditions: nominal → cold → high wind → sensor drift, with observable changes in compensation mode, control parameters, health state and tracking behavior.

4. System Architecture

Layer	Main elements	Role
Sensing	EO camera, temperature, pressure, humidity, wind, IMU, optional dust sensing	Observe target and environment
Edge AI	YOLO-based detector, tracker, performance model	Extract target information and estimate state
Environmental engine	Condition classification and compensation logic	Convert environmental state into control/health adjustments
Real-time control	MCU, motor control, encoder	Execute deterministic actuation and

Layer	Main elements	Role
	feedback, heater control	protection
Actuation	Pan axis, tilt axis, thermal actuator	Stabilize sensor and maintain operating temperature
Health & dashboard	Health monitor, Streamlit UI, alerts	Expose system condition and operator information

HAADS HIGH-LEVEL DATA & CONTROL FLOW
EO CAMERA → AI DETECTION → TARGET TRACKING
↓
ENVIRONMENT SENSORS → ENVIRONMENT ENGINE → ADAPTIVE COMPENSATION
↓
PAN-TILT + THERMAL CONTROL
↓
HEALTH MONITOR → DASHBOARD

High-altitude sensing concept image from the supplied source document.

5. Detection, Identification & Tracking

5.1 EO vision pipeline

The source document proposes an EO camera pipeline of Camera → Video Frames → AI Detection Model → Candidate → Tracking Algorithm → Pan-Tilt Command. It also discusses drone/bird/aircraft/background discrimination as a future classification capability. ṆfileciteÖturn5file5ÖL688-L737Ộ

CURRENT SOFTWARE-ALIGNED VISION FLOW
CAMERA / VIDEO INPUT
↓
YOLO26n / SUITABLE DETECTOR

↓
OBJECT BOX + CONFIDENCE
↓
TRACKER: POSITION / MOTION STATE
↓
PAN-TILT ERROR
↓
CONTROL COMMAND

CURRENT PROTOTYPE STATUS

The current camera demonstration uses YOLO26n for general object detection. Do not describe the generic model as a validated drone detector. Drone-specific identification should be claimed only after a drone-trained/fine-tuned model and evaluation dataset are integrated.

5.2 RF monitoring concept

The source proposes RF detection as an optional sensing subsystem using an SDR receiver, antenna, signal processing and spectrum-monitoring software. Any RF implementation in this project is scoped to receive-only sensing and must follow applicable spectrum regulations. The placeholder values such as “XXXX MHz” and “XX dBm” from the source are intentionally removed from this final document because they are not actual measured project values. ṆfileciteÖturn5file9ÖL1114-L1115Ộ

5.3 Thermal sensing concept

A thermal camera is identified as a complementary option for low-light, night and poor-visibility conditions. For the student prototype, this can remain simulated or future hardware because the source document explicitly allows simulation when hardware cost is high. ṆfileciteÖturn5file5ÖL706-L711Ộ

6. Precision Pan-Tilt Tracking

The camera/sensor is mounted on a two-axis pan-tilt mechanism. The controller converts target image position into pan and tilt error and then drives the corresponding actuators. The source identifies pan-tilt tracking as one of the most important parts of the proposed system. ṆfileciteÖturn5file8ÖL937-L952Ộ

Signal	Origin	Control use
Target X/Y	Vision detector/tracker	Calculate image-center error
IMU attitude/vibration	IMU	Estimate platform disturbance
Wind estimate	Wind sensor / simulated input	Increase stabilization effort
Encoder position	Motor/gimbal feedback	Close the position loop
Temperature	Environmental sensor	Compensate mechanical/control behavior

7. Temperature-Dependent Cable Rigidity Compensation

At low temperature, cable assemblies can become stiffer and less flexible. If the cable routing exerts resistance on a pan-tilt axis, it can introduce unwanted torque, increasing pointing error, motor load and oscillation risk. This mechanism is explicitly described in the source document. ṆfileciteÖturn5file8ÖL953-L988Ộ

CABLE COMPENSATION CONCEPT
MEASURE TEMPERATURE
↓
TEMPERATURE-DEPENDENT CABLE STIFFNESS MODEL
↓
ESTIMATE DISTURBANCE TORQUE
↓
ADD COMPENSATION TO MOTOR COMMAND
↓
CHECK POSITION / CURRENT FEEDBACK

7.1 Conceptual model

A simplified conceptual relationship is $\tau c \approx k(T) \cdot \theta$, where τc is estimated compensation torque, $k(T)$ is temperature-dependent effective stiffness and θ is the relevant angular displacement. This is a model concept, not a validated mechanical identification result. The actual function should be identified experimentally from the selected cable routing, material and gimbal geometry.

7.2 Validation plan

- Measure axis error with flexible/nominal cable routing.
- Repeat at controlled temperature points or with a safe thermal simulation.
- Record motor effort/current and position error.
- Fit an empirical temperature-to-disturbance model.
- Compare tracking behavior with compensation disabled and enabled.

8. Wind Disturbance Compensation

Strong wind can disturb the camera and pan-tilt assembly. The proposed solution combines IMU measurements, wind information and motor encoder feedback so that the controller can estimate disturbance and correct pointing angle. ṆfileciteÖturn5file8ÖL989-L1004Ộ

WIND COMPENSATION LOOP
WIND DISTURBANCE
↓
IMU + WIND SENSOR + ENCODER
↓
ESTIMATE ATTITUDE / POSITION ERROR
↓
ADAPTIVE CONTROL
↓
PAN-TILT CORRECTION

8.1 Control behavior

At increasing wind levels, the controller can transition through operating modes such as NORMAL → HIGH-WIND → EXTREME-WIND. The exact thresholds should be configuration parameters validated during testing rather than treated as universal limits.

9. Adaptive Control System

Instead of keeping fixed controller parameters for every environment, the proposed architecture uses temperature, wind and pressure as context inputs to an adaptive controller. The source explicitly contrasts a fixed PID loop with an adaptive arrangement that changes parameters with environmental conditions. ṆfileciteÖturn5file4ÖL517-L538Ộ

Environment state	Possible controller response	Purpose
Nominal	Baseline gains / normal stabilization	Smooth tracking
Cold	Increase modeled disturbance compensation; monitor stiffness-related load	Counter temperature-induced resistance
High wind	Increase disturbance rejection / stabilization mode	Reduce pointing disturbance
High vibration	Increase filtering and health scrutiny	Avoid reacting to noise as target motion
Sensor drift warning	Apply calibrated correction / reduce confidence	Prevent biased feedback

10. Sensor Drift Compensation

The source document identifies temperature-related sensor drift as a potential source of error and proposes using calibration data to map measured temperature to a corrected sensor value. ṆfileciteÖturn5file4ÖL539-L553Ộ

SENSOR DRIFT CORRECTION
MEASURED SENSOR VALUE
+
MEASURED TEMPERATURE

↓
CALIBRATION TABLE / MODEL
↓
CORRECTED SENSOR VALUE
↓
CONTROL + HEALTH MONITOR

ENGINEERING NOTE

The source includes an illustrative IMU example with a +1.5° cold-environment error. That value is retained only as an example of the concept, not as a measured characteristic of the prototype.

11. Environmental Monitoring Subsystem

The source identifies temperature, atmospheric pressure, humidity, wind speed, vibration and optional dust ingress as environmental parameters. The purpose is to estimate how much the environment is affecting system performance. ṆfileciteÖturn5file0ÖL56-L70Ọ

Parameter	Prototype / proposed sensor	Used by
Temperature	Digital temperature sensor / BME280 for simulation	Thermal control, cable compensation, drift model, health
Pressure	Barometric pressure sensor / BME280 for simulation	Environmental state and performance model
Humidity	Humidity sensor / BME280 for simulation	Moisture/condensation risk
Wind	Anemometer or simulated wind input	Disturbance compensation
Vibration / attitude	IMU / MPU6050 for simulation	Stabilization and health
Dust ingress	Optional particulate / ingress sensor	Ruggedization and maintenance risk

12. Thermal Management

The source proposes insulated enclosure construction, controlled heaters, thermal pads, heat spreaders, temperature monitoring and low-temperature-rated components. It also notes that lower air density can reduce the effectiveness of ordinary air cooling. ŅfileciteÖturn5file4ÖL1043-L1064Ọ

CLOSED-LOOP THERMAL CONTROL
EXTERNAL / INTERNAL TEMPERATURE
↓
TEMPERATURE CONTROLLER
↙️ ↘️
HEATER MONITORING
↓
SAFE OPERATING RANGE
↓
REDUCE HEATER POWER

12.1 Heater control concept

The recommended architecture is closed-loop control rather than permanently energizing a heater. A future hardware implementation can use an MCU GPIO → MOSFET driver → power MOSFET → heater element chain with temperature/current limits and an independent hardware safety cutoff. This follows the source document's proposed design approach.

13. Battery Performance & Power Health

Cold conditions can reduce available battery capacity, cause voltage drop and increase internal resistance. The proposed battery-management architecture monitors voltage, current and temperature and feeds those values into battery-health prediction. ŅfileciteÖturn5file3ÖL362-L382Ọ

Measurement	Why it matters	Dashboard output
Voltage	Detect sag / supply risk	Power status

Measurement	Why it matters	Dashboard output
Current	Estimate load and motor demand	Load / overload warning
Battery temperature	Identify cold stress	Thermal warning
Estimated capacity	Assess remaining usable energy	Battery health / runtime estimate

CLAIM CONTROL

The source contains example battery values (for example, 82% battery and 68% estimated capacity at −10°C). These are not presented as measured project results.

14. Ruggedized Mechanical & Environmental Design

The source proposes protecting the enclosure against dust, snow, moisture, vibration and cold, using an aluminium-alloy enclosure, sealed connectors, protective coating, low-temperature-rated bearings, appropriate lubrication, vibration isolators, reinforced mounting, conformal coating and wide-temperature components. ṆfileciteÖturn5file3ÖL383-L402Ộ

Design area	Proposed treatment
Enclosure	Aluminium alloy / rugged housing concept
Ingress protection	Sealed connectors, protective interfaces
Mechanical	Low-temperature bearings, suitable lubrication, reinforced mounting
Vibration	Isolation and mechanical damping
Electronics	Conformal coating where appropriate, wide-temperature components
Thermal	Insulation + controlled heating + monitored temperature

15. System Health Monitoring

The system health monitor is intended to continuously report the condition of key subsystems. The source lists CPU temperature, battery health, motor current, sensor status, camera status, RF receiver status and communication status. ṆfileciteÖturn5file3ÖL403-L416Ộ

Health signal	Example state	Action
CPU / AI temperature	NORMAL / WARNING	Reduce load or alert operator
Battery	NORMAL / WARNING / CRITICAL	Power-risk alert
Motor current	NORMAL / HIGH	Inspect load / compensation
Sensor communication	OK / LOST	Flag degraded subsystem
Camera	OK / LOST	Stop tracking / alert
Communication	OK / LOST	Operator warning
Thermal controller	ON / OFF / FAULT	Protect electronics

16. Predictive Performance Assessment

The source proposes predicting whether performance may degrade using temperature, wind, pressure, battery condition, motor current, sensor drift and vibration. Candidate algorithms include Random Forest, XGBoost and neural networks, with Random Forest suggested as a simpler initial prototype. ṆfileciteÖturn5file3ÖL417-L446Ộ

PREDICTIVE PERFORMANCE PIPELINE
TEMPERATURE • WIND • PRESSURE • BATTERY • MOTOR CURRENT • DRIFT • VIBRATION
↓
PERFORMANCE MODEL
↓
RISK / DEGRADATION SCORE
↓

OPERATOR STATUS + HEALTH WARNING

16.1 Recommended dataset structure

Input	Example representation	Target
Temperature	°C	Tracking error / health class
Wind speed	km/h or m/s	Tracking error / stabilization state
Pressure	hPa	Environmental state
Battery	Voltage/current/temp/estimated capacity	Power-health class
Motor current	A	Mechanical load
Vibration	IMU features	Stability class
Tracking error	degrees / pixels	Measured outcome

17. Software Architecture & Data Flow

The upgraded architecture aligns the source document’s conceptual pipeline with the current software structure: camera input, AI detection, tracking, environmental engine, compensation, health monitoring, performance assessment and Streamlit visualization.

Module	Responsibility
camera.py	Camera acquisition and stream lifecycle
detector.py	YOLO26n inference / general object detection
tracker.py	Target state and tracking logic
environment.py	Environmental state and simulation inputs
compensation.py	Environmental compensation rules/models

Module	Responsibility
performance.py	Performance/degradation assessment
health_monitor.py	Subsystem health state
hardware_interface.py	Wokwi / embedded telemetry interface concept
data_manager.py	Shared system state / persistence
dashboard.py	Dashboard components
app.py	Application orchestration
system_data.json	Current system state / telemetry representation

CURRENT SOFTWARE PROTOTYPE FLOW
CAMERA
↓
YOLO26n DETECTION
↓
OBJECT LOCATION + CONFIDENCE
↓
TARGET TRACKING
↓
ENVIRONMENT ENGINE
↓
COMPENSATION + PERFORMANCE

↓
SYSTEM HEALTH
↓
STREAMLIT LIVE DASHBOARD

18. Wokwi Hardware Simulation

Wokwi is used as the simulation-first embedded layer so the control concept can be demonstrated before physical components are purchased. The current simulation architecture uses an ESP32, BME280, MPU6050, pan servo, tilt servo and potentiometer-driven environmental inputs.

Simulated element	Interface / role
ESP32	Embedded controller / telemetry
BME280	Temperature, pressure and humidity
MPU6050	Acceleration / IMU feedback
Pan servo	Pan-axis actuator concept
Tilt servo	Tilt-axis actuator concept
Potentiometer inputs	Controlled temperature / pressure / wind / vibration scenarios

BME280 reference image from the supplied project document.

SIMULATION STATUS

Wokwi validates the embedded control concept and I/O behavior. It does not constitute field validation of high-altitude mechanical, thermal, RF or environmental performance.

19. Ground-Control Dashboard

The Streamlit dashboard is the operator-facing layer for system status, camera output, environmental state, compensation state, virtual pointing, performance and subsystem health. The source document proposes a dashboard with environmental readings, tracking state, compensation status and health indicators. ṆfileciteÖturn5file6ÖL1312-L1335Ộ

Dashboard panel	Recommended content
System Overview	Operational state, timestamp, major alerts
Live Camera / AI	Video, bounding boxes, object count, confidence
Environment	Temperature, pressure, humidity, wind, vibration
Compensation	Active modes, control gains/parameters, correction state
Virtual Pointing	Pan/tilt command, target error, simulated actuator state
Performance	Risk/degradation score, trends, test scenario
Hardware Simulation	Wokwi connection state and telemetry
Health	CPU/AI, camera, sensors, communications, power
Alerts	Cold, high wind, drift, high load, communication fault

DEMO LANGUAGE

Use “Object Detection & Tracking” for the current generic YOLO26n camera demo. Use “Drone Detection” only when a drone-specific model has actually been trained/fine-tuned and evaluated.

20. Proposed Hardware Architecture

The source recommends separating computationally intensive AI processing from deterministic real-time control. A proposed architecture uses an edge AI computer, a real-time MCU/control PCB, EO camera, pan-tilt with encoder feedback, environmental sensors, protected power, thermal control and UART/CAN communication. ṆfileciteÖturn5file2ÖL278-L291Ộ

Subsystem	Proposed architecture	Status
AI compute	Raspberry Pi CM5 + Hailo AI accelerator (proposed option)	Future hardware
Real-time control	STM32H7 series (proposed)	Future hardware
Vision	EO camera	Current camera demo / future rugged camera
Tracking	Encoder-feedback pan-tilt	Concept / Wokwi control simulation
Environment	Temperature + pressure + humidity + wind	Wokwi/simulated; future physical sensors
Stabilization	IMU-based feedback	Wokwi MPU6050 simulation
Thermal	Closed-loop heater control	Concept / future hardware
Communication	UART + CAN; Ethernet where required	Architecture proposal
Dashboard	Streamlit	Current software prototype

Edge-computing form-factor reference image retained from the source document.

21. High-Altitude Environmental Control PCB Concept

The source proposes a custom control PCB with power management, STM32 MCU, environmental sensors, motor control, heater control and AI interface. It recommends a 4-layer PCB with signal/component, solid ground, power and bottom-signal layers. ṆfileciteÖturn5file6ÖL801-L827Ộ

PROPOSED CONTROL PCB FUNCTIONAL FLOW
POWER INPUT
↓
POWER MANAGEMENT / PROTECTION
↓
STM32H7 REAL-TIME CONTROLLER
↙ ↓ ↘
SENSORS MOTOR CONTROL HEATER CONTROL

↓ ↓ ↓
I²C / SPI PWM / ENCODER MOSFET + FEEDBACK
↘ ↓ ↙
AI INTERFACE / UART / CAN / ETHERNET

21.1 PCB sections

8. Power management: input protection, fuse/protection, reverse-polarity protection, surge protection and DC-DC conversion.
9. MCU: STM32, clock/reset, SWD and decoupling.
10. Sensor section: temperature, pressure, humidity and IMU; keep IMU away from high-current and noisy switching areas.
11. Motor control: driver interface, encoder input, current sensing and protection.
12. Heater control: MOSFET/gate control, temperature feedback and heater connector.
13. Communication: UART, CAN and Ethernet as required.

22. Communication & Telemetry Architecture

Link	Purpose	Typical data
AI → MCU	Target/control commands	Target X/Y, confidence, tracking mode
MCU → AI	Telemetry	Environment, actuator state, health
I²C	Environmental sensors	Temperature, pressure, humidity
SPI	High-speed sensors where required	IMU / sensor data
PWM	Actuator command	Pan/tilt or heater control
CAN	Rugged distributed subsystem link	Health/command telemetry
Ethernet	High-rate network/video path where required	Video / system data

The source gives an example AI-to-MCU message containing target X, target Y, confidence and tracking mode, followed by coordinate conversion into pan and tilt error. ṆfileciteÖturn5file6ÖL1133-L1154Ộ

23. Environmental Compensation Engine

This engine is the core innovation layer. The source lists temperature, pressure, wind, IMU, motor current, encoder and battery as inputs, with outputs including motor-gain adjustment, sensor-drift correction, thermal control, performance prediction and health warning. ṆfileciteÖturn5file6ÖL1156-L1179Ộ

Input	Derived state	Compensation / output
Temperature	Cold / nominal / thermal risk	Cable stiffness model, control adjustment, heater state
Wind	Normal / high / extreme	Stabilization mode / gain adjustment
Pressure	Environmental state	Performance context
IMU	Attitude / vibration	Stabilization and drift monitoring
Motor current	Load estimate	Health warning / compensation evidence
Encoder	Actual axis position	Closed-loop position correction
Battery	Power-health state	Runtime / health warning

23.1 Rule-based prototype logic

A simple first implementation can use thresholded rules: if temperature is below a configured low-temperature threshold, enable cold compensation; if wind exceeds a configured threshold, enable stabilization mode; if estimated drift exceeds a limit, apply sensor correction; if motor current exceeds a limit, generate a health warning. The source explicitly proposes this style before replacing fixed rules with adaptive PID, Kalman filtering or machine-learning prediction. ṆfileciteÖturn5file6ÖL1181-L1197Ộ

24. Implementation Plan & Roadmap

Phase	Work package	Exit criterion
1	Simulation foundation	Environment + tracking scenarios run repeatably
2	AI vision	Camera + detector + tracking

Phase	Work package	Exit criterion
		pipeline operational
3	Environmental sensing	Temperature/pressure/IMU and other inputs integrated
4	Pan-tilt prototype	Target error drives simulated/physical axis commands
5	Compensation	Cold, wind and drift compensation demonstrated
6	Health + prediction	Fault states and degradation score visible
7	Hardware integration	Physical sensors/MCU integrated after software validation
8	Custom PCB	Schematic/layout after subsystem validation
9	Controlled validation	Repeatable before/after test data collected

This sequence consolidates the source document’s implementation plan, which starts with simulation, then AI detection, environmental sensors, pan-tilt, compensation and finally custom PCB integration. ṆfileciteÖturn5file6ÖL1337-L1379Ộ

25. Validation & Test Plan

25.1 Scenario matrix

Scenario	Inputs	Observe	Expected engineering behavior
Nominal	Moderate temperature, low wind	Tracking + health	Baseline behavior
Extreme cold	Low temperature	Compensation + motor effort + health	Cold mode activates; disturbance model changes

Scenario	Inputs	Observe	Expected engineering behavior
High wind	High wind input	Pointing error + stabilization	Adaptive stabilization activates
Sensor drift	Injected bias	Corrected sensor value	Calibration correction reduces bias
High vibration	Injected vibration	Tracker/control response	Filtering and health warning respond appropriately
Power stress	Reduced voltage / increased load	Battery and current health	Warning state appears before unsafe condition
Communication loss	Disconnect telemetry path	Health monitor	Subsystem fault reported

25.2 Metrics to record

- Detection latency / inference rate
- Tracking continuity and target-loss events
- Image-space or angular tracking error
- Pan/tilt settling behavior
- Motor command and current proxy
- Environmental state
- Compensation mode
- Health state transitions
- False-alarm behavior on representative non-target objects
- Power/thermal state

25.3 Before-vs-after comparison

The strongest demonstration should compare the same environmental scenario with compensation disabled and enabled. The source specifically recommends a before/after compensation demonstration for extreme cold and high wind. ṆfileciteÖturn5file1ÖL171-L190Ộ

26. Current Prototype Status & Evidence Policy

Area	Current status	How to present it
Streamlit dashboard	Implemented	Show live dashboard screenshots
Camera + YOLO26n	Implemented as general object	Say “general object detection &

Area	Current status	How to present it
	detection	tracking”
Environmental simulation	Implemented	Show temperature/pressure/wind scenario changes
Compensation engine	Implemented as prototype logic	Show mode/parameter changes
System health	Implemented in dashboard	Show subsystem states and alerts
Wokwi	Implemented as embedded simulation concept	Show circuit and servo/sensor behavior
Physical hardware	Not yet validated as a complete deployed unit	Describe as proposed/future
Drone-specific AI	Not established by generic model alone	Mark as future fine-tuning/evaluation
High-altitude field validation	Not established	Do not claim field performance

WHAT COUNTS AS EVIDENCE

A screenshot proves the software/simulation state visible in it; it does not prove physical high-altitude performance. Keep screenshots, logs, configuration values and test datasets alongside measured results in the project repository.

27. Limitations & Risk Register

Risk / limitation	Impact	Mitigation
Generic detector not drone-specific	False target classification	Fine-tune/evaluate a drone dataset before claiming drone detection
No physical high-altitude test yet	Thermal/mechanical claims unverified	Use simulation + controlled chamber/wind tests before field trials
Cable model is conceptual	Compensation may be inaccurate	Identify model experimentally

Risk / limitation	Impact	Mitigation
Wind input may be simulated	Real disturbance may differ	Integrate calibrated wind sensing and controlled tests
Sensor drift coefficients unknown	Correction may be biased	Temperature-point calibration
Battery model not measured	Runtime prediction uncertainty	Collect voltage/current/temperature data
RF subsystem not implemented as a validated detector	No evidence of RF detection performance	Keep RF receive-only and future/optional
Wokwi differs from hardware physics	Simulation-to-hardware gap	Validate each subsystem on physical hardware before integration

28. Safety, Compliance & Prototype Scope

The project is framed as an academic sensing, tracking, environmental-compensation and health-monitoring system. The safe prototype scope in the source is Detection + Identification + Tracking + Environmental Compensation + Health Monitoring. For the response stage, the source recommends only alert generation, operator notification, tracking handoff and a simulated authorized response workflow. ṆfileciteÖturn5file1ÖL135-L143Ộ

SCOPE BOUNDARY

This documentation does not specify RF jamming, communication disruption, signal interception, weapon control or physical neutralization. Any future operational system would require appropriate legal, spectrum, aviation, safety and organizational authorization.

29. Prototype BOM — Simulation First, Hardware Later

Item	Role	Current state	Future note
ESP32	Embedded control simulation	Wokwi	Physical MCU optional for first prototype
BME280	Temperature / pressure / humidity simulation	Wokwi	Physical environmental sensor
MPU6050	IMU simulation	Wokwi	Physical IMU / better-

Item	Role	Current state	Future note
			grade IMU later
Pan servo	Pan-axis simulation	Wokwi	Encoder-feedback actuator for rugged prototype
Tilt servo	Tilt-axis simulation	Wokwi	Encoder-feedback actuator for rugged prototype
Potentiometers	Controlled environment inputs	Wokwi	Replace with physical wind/other sensors
EO camera	Vision input	Current software prototype	Rugged camera / low-light or thermal option later
Edge AI computer	AI inference	Proposed	Raspberry Pi CM5 + Hailo is one proposed architecture
STM32H7	Real-time control	Proposed	Future control PCB
Custom PCB	Integration	Proposed	After subsystem validation

30. Final Project Deliverables

14. System architecture and block diagrams
15. Environmental impact and compensation analysis
16. EO AI detection and tracking prototype
17. Environmental monitoring and simulation
18. Pan-tilt control concept
19. Adaptive compensation engine
20. System health monitor
21. Predictive performance model / risk score
22. Streamlit ground-control dashboard
23. Wokwi embedded simulation
24. Hardware architecture and BOM
25. PCB concept and integration roadmap
26. Validation plan and measured test results
27. Project source code and reproducibility documentation

31. Glossary

Term	Meaning
HAADS	High-Altitude Anti-Drone System
EO	Electro-Optical camera/sensing
IMU	Inertial Measurement Unit
MCU	Microcontroller Unit
BMS	Battery Management System
PID	Proportional–Integral–Derivative control
RF	Radio Frequency
SDR	Software-Defined Radio
NPU	Neural Processing Unit / AI accelerator
CAN	Controller Area Network
I^{2}C	Inter-Integrated Circuit
SPI	Serial Peripheral Interface
PWM	Pulse-Width Modulation
SIH	Smart India Hackathon
PS ID	Problem Statement Identifier

32. References & Source Basis

This document is an upgraded engineering edition of the supplied “Indreasena Project.docx”. The source document establishes the problem framing, proposed architecture, environmental compensation concepts, hardware recommendations, implementation phases and safe prototype scope. Relevant source excerpts are preserved and reorganized rather than silently replaced. ṆfileciteÖturn5file0ÖL19-L28Ộ ṆfileciteÖturn5file2ÖL265-L291Ộ

Reference	Use in project
Supplied project document — Indreasena Project.docx	Primary project concept, architecture, compensation methods, hardware proposal and implementation roadmap
Ultralytics YOLO26 documentation	AI model family / detection implementation reference
Wokwi ESP32 documentation	Embedded simulation and ESP32 peripheral reference
Raspberry Pi AI Camera documentation	Future edge-AI camera architecture reference
Smart India Hackathon PS 26050	Problem-statement context and competition framing

Recommended online references for the final repository should be kept as clickable hyperlinks in the project README. Use the official documentation pages rather than third-party summaries.

- https://docs.ultralytics.com/models/yolo26
- https://docs.wokwi.com/guides/esp32
- https://docs.wokwi.com/guides/esp32-wifi
- https://www.raspberrypi.com/documentation/accessories/ai-camera.html
- https://sih.dsce.in/sih

33. Appendix A — Recommended Demonstration Script

28. Start the Streamlit dashboard and show system overview.
29. Show the live camera with general object detection and tracking.
30. Open the environment panel and set a nominal condition.
31. Increase wind input and show stabilization/compensation state changing.
32. Lower temperature and show cold compensation / thermal status.
33. Inject a sensor-drift condition and show corrected state.
34. Open Wokwi and show ESP32 sensor/servo simulation.
35. Show health-monitor status and generated alerts.
36. Explain which functions are current prototype evidence and which are future hardware capabilities.
37. Conclude with the environmental-aware feedback loop: sense → estimate → compensate → stabilize → monitor.

34. Appendix B — Recommended Repository Structure

A reproducible repository should separate application code, embedded simulation, hardware design and documentation. A recommended structure is:

```
HAADS_SIH_26050/
├── README.md
├── app/
│   ├── app.py
```

```
│   ├── detector.py
│   ├── tracker.py
│   ├── environment.py
│   ├── compensation.py
│   ├── performance.py
│   ├── health_monitor.py
│   ├── hardware_interface.py
│   ├── data_manager.py
│   └── dashboard.py
├── prototype/
│   └── wokwi/
│       ├── diagram.json
│       ├── wokwi.ino
│       └── libraries.txt
├── models/
├── data/
├── hardware/
│   ├── BOM.md
│   ├── wiring.md
│   └── pcb/
├── docs/
│   ├── architecture.md
│   ├── compensation.md
│   ├── testing.md
│   └── roadmap.md
└── results/
    ├── screenshots/
    └── test-results/
```

35. Appendix C — Engineering Claim Checklist

Claim	Allowed wording now	Evidence needed for stronger claim
AI detection	YOLO26n general object detection	Drone dataset + metrics
Tracking	Target/object tracking in prototype	Controlled tracking accuracy study
Environmental compensation	Prototype compensation logic / simulation	Physical environmental test data
Cable compensation	Conceptual temperature-dependent model	Mechanical characterization
Wind compensation	Simulation/control concept	Controlled wind disturbance test
Thermal management	Proposed closed-loop architecture	Thermal chamber / cold test

Claim	Allowed wording now	Evidence needed for stronger claim
Battery health	Monitoring/prediction concept	Battery test dataset
High-altitude robustness	Design objective / simulation concept	Altitude-representative validation
Mission readiness	Do not report unsupported percentage	Validated system-level test campaign

FINAL ENGINEERING POSITION

The project is strongest when it clearly separates what is already demonstrated in software/Wokwi from what is proposed for future physical validation. This upgraded document is structured around that distinction.

HAADS • SIH 26050

Sense the environment. Compensate the disturbance. Stabilize the track. Monitor system health.